Controller Semantics and Negative Results in a Synthetic Cross-Layer Orchestration Study

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gunnchOS Research Exploratory Preprint — v0.1
Local candidate for scientific review, 14 October 2026
Not peer reviewed. Not posted. Publication authorization: false.

Abstract

Digital-twin context can guide network and computation choices, but a controller label does not establish information equivalence, meaningful optimization, or improved service. This article audits two preserved synthetic cross-layer orchestration implementations and their negative results. In the original five-seed held-out study, the procedural twin policy underperforms cloud fallback by 0.01360 utility units, with a post hoc paired 95% interval [-0.01478,-0.01242]. A separately reproduced revision retains a negative difference of 0.01255 and unfavorable adaptive-fidelity and checkpoint outcomes. Source inequalities show that the full linear selector chooses edge placement on the available path with the smallest configured penalty; focused assertions confirm this behavior on 9,600 reconstructed input states in each version, separately. A zero-latency transformation instead creates score ties. The symmetric SLSQP share objective has a uniform mathematical optimum, but optimizes a different function from network/placement utility and returns small budget residuals. A bounded numerical analysis finds these residuals too small to reverse the retained negative comparisons. Switching accounting, incomplete action feasibility, unequal consumed information/computation, and truth-changing interventions limit causal interpretation. The contribution is a reproducible methodological characterization of this model, with no measured network, general adaptive superiority, or established novelty claim. **Keywords:** Cross-layer control, controller semantics, digital twin, switching costs, negative results, reproducibility.

1 Introduction

Service management across network paths and device, edge, and cloud computation requires more than choosing the configuration with the lowest instantaneous delay. Changes themselves can consume resources, interrupt work, or create unstable operation. A controller may improve an instantaneous quality measure while reducing the value accumulated over a sequence of decisions. The IMT-2030 framework motivates resilient future communication systems [5]; it does not validate a controller or provide physical measurements for this study.

Mobile edge computing research already treats communication and computation as coupled decisions [8]. Digital-twin-assisted offloading further uses virtual representations to estimate state, support decisions, or train policies [2, 3, 9]. These precedents establish a substantive novelty boundary: using network context to select an edge server is established work. A twin label does not identify an algorithmic contribution, establish equal baseline information, or demonstrate an improvement in deployed service.

This paper examines a narrower question: *what conclusions survive reproduction and source-level audit of a preserved, twin-informed synthetic orchestration experiment when switching costs and baseline semantics are made explicit?* The implementation combines network selection, placement, spectrum shares, and a power proxy. It originally hypothesized that a procedural twin policy would improve continuity over a baseline named `no_adaptation`. Reproduction with the required optimizer preserves the contrary result. Correcting that claim is part of the scientific result.

First, we preserve the existing exact numerical reproduction of comparable non-timing outputs from two source revisions and retain separate protocols, seed lists, and generators. Second, we quantify the original negative comparison with seed-paired uncertainty and decompose gross continuity and switching penalties without retuning. Third, source inspection identifies where baseline labels, information equivalence, and ablation terminology exceed enforcement. Finally, separately checked source inequalities explain the linear selector's effective behavior, distinguish an exact share optimum from numerical residuals, and define a prospective fair comparison. The reproduced extension retains separate unfavorable fidelity/checkpoint outcomes.

The contribution is an implementation-specific methodological case study. It is not a new globally optimal controller, a learned network digital twin, or an assessment of every twin architecture. All orchestration evidence is synthetic. The present contribution is

selected as a methodological investigation, rather than a fair causal demonstration of controller superiority. There is no calibrated RF channel, packet transmission, measured handover interruption, carrier experiment, or physical edge-device energy measurement. Competition and companion radio projects are context, not quantitative evidence for these results.

2 Related Work and Novelty Boundary

Mao *et al.* survey joint radio and computation management [8]. Their energy-harvesting offloading study includes dynamic energy state and task execution cost [7]. Our model instead uses an independently drawn energy-budget covariate and an uncalibrated formula labeled in joules. It does not evolve a battery or constrain depletion. The literature motivates the problem, without validating our proxy.

Sun *et al.* investigate latency reduction in digital-twin edge networks [9]. Digital-twin-assisted offloading is therefore established problem context. Dai *et al.* combine stochastic offloading, Lyapunov reformulation, and actor-critic learning [2]. Do-Duy *et al.* optimize edge selection and offloading under latency and resource constraints [3]. Our frozen linear and procedural rules neither implement nor compare against reproduced versions of these methods. No performance claim against them follows from our tables.

Hui *et al.* consider digital-twin-enabled collaboration for composite vehicular services [4]. Their stated scope is composite vehicular services, distinct from the categorical continuity flag used here. A 2026 preprint by Almohammedi *et al.* studies fidelity-aware arbitration between competing Open RAN applications [1]. It monitors prediction discrepancy and uses fallback. This is a close conceptual competitor for trustworthy twin-assisted control. Its twin fidelity means predictive agreement; our extension uses *service fidelity* as a synthetic utility multiplier. These are distinct quantities. The source is cited as a preprint; acceptance is not assumed.

Prediction and transition-aware service placement also precede this study. Wang *et al.* [10] model local and migration costs over a look-ahead window and examine prediction error; their author manuscript reports synthetic and mobility-trace evaluation. Our current-slot rules do not implement that horizon optimization or reproduce its performance. A lower paired utility cannot establish a new switching-control principle. Jain *et al.* [6] supply the established resource-allocation index used here; fairness of shares is not fairness of delivered task service. These distinctions constrain both algorithmic and methodological novelty.

Our distinction is the reproducible audit and mathematical characterization of a specific archived comparison: a preserved negative hypothesis, paired inference, switching decomposition, and a diagnosis of accessible features versus consumed features and computational work. We do not claim the first twin orchestrator, switching-aware offloading model, or robust fallback mechanism. The study is not a field-valid benchmark. A publication decision should assess the usefulness of this bounded methodological contribution.

3 System and Analytical Model

3.1 State, Actions, and Synthetic Twin Context

At slot t, state s_t contains base latency ℓ_t, loss percentage q_t, spectrum budget B, energy covariate e_t, user count N_t, priorities, outage, available networks, node capacities, mobility, blockage, and continuity class. An action is

$$a_t = (\boldsymbol{x}_t, \boldsymbol{p}_t, n_t, v_t), \tag{1}$$

where x_{ti} is a user's spectrum share, p_{ti} a dimensionless power setting, n_t a network category, and v_t placement. Networks include terrestrial, local-edge Wi-Fi, degraded-local, and NTN fallback; placement is cloud, edge, or local. These names select coefficients rather than implement transport protocols or hardware.

The synthetic twin is structured context plus an analytical predictor. There is no observation stream synchronized to a physical twin, model fitting, calibration error process, or deployed feedback. Mobility and blockage are recorded covariates. The original procedural policy does not directly use them and the score is independent of them. Radio context describes an intended abstraction, not measured radio awareness.

Define aggregate utilization

$$u_t = \mathrm{clip}_{[0,1.5]}\left(\frac{\sum_i x_{ti}}{B}\right). \tag{2}$$

Network multipliers α_n are 1.0, 1.1, 1.6, 2.2, 1.4, and 10.0 for terrestrial, local-edge Wi-Fi, degraded-local, NTN fallback, device-to-device, and offline continuation. Placement multipliers β_v are 1.15, 1.0, and 1.25 for cloud, edge, and local. The original generator uses the first four networks. Its score is

$$\widehat{L}_t = \ell_t \alpha_{n_t} \beta_{v_t} (1 + 0.4u_t), \tag{3}$$

$$\widehat{q}_t = \min\{100, q_t \alpha_{n_t}(1 + 0.2u_t)\}, \tag{4}$$

$$r_t = \mathrm{clip}_{[0,1]}(1 - \widehat{q}_t/100), \tag{5}$$

$$g_t = \mathrm{clip}_{[0,1]}\left[r_t\left(1 - \frac{\min\{\widehat{L}_t, 500\}}{500}\right)\right]. \tag{6}$$

The gross continuity g_t uses a configured 500 ms normalization, not an application-validated deadline. Increasing utilization increases modeled latency, with no throughput or task-completion benefit from additional shares. This limits the meaning of spectrum optimization.

3.2 Switching, Energy, and Fairness

The transition charge is

$$c_t = 0.05\mathbb{1}[n_t \neq n_{t-1}] + 0.03\mathbb{1}[v_t \neq v_{t-1}], \tag{7}$$

with $c_1 = 0$. Net utility is $U_t = \max\{-1, g_t - c_t\}$. Charges are dimensionless deductions, not measured

transition durations or energy. Episode utility is the mean of $T = 16$ slots, whereas reported charge C is the episode sum. Thus C/T is comparable with average utility.

The energy proxy is

$$\widehat{E}_t = (0.2e_t + 15u_t + 0.05\widehat{L}_t)(0.6 + 0.4\bar{p}_t), \quad (8)$$

where $\bar{p}_t = N_t^{-1} \sum_i p_{ti}$. Output keys label it joules, but no device model or calibration supports that unit. The covariate e_t is not a battery balance and (8) does not enforce an energy constraint. Jain's expression $J_t = (\sum_i x_{ti})^2/(N_t \sum_i x_{ti}^2)$ describes share equality, not observed service fairness. Continuity depends on the sum of shares; distribution changes can affect J_t without affecting continuity. These metrics do not supply independent physical validations.

3.3 Feasibility Boundary

Several policies enumerate available categories and positive-capacity placements, but scoring does not reject unavailable networks or over-capacity placement. Capacity mostly acts as a selection flag rather than a workload-dependent server limit. A static action can survive a later outage and still receive a finite score based on its category multiplier. Reported utility evaluates declared actions under the formula, not necessarily feasible service. This defect remains in the historical reproduction.

4 Implemented Controllers and Audit

4.1 Procedural Policy and Share Optimization

The procedural twin policy selects available local-edge Wi-Fi during outage, then degraded-local or NTN fallback. Outside outage it selects Wi-Fi when base latency exceeds 40 ms; otherwise it prefers terrestrial. Placement becomes edge for strict continuity or latency above 40 ms with positive edge capacity; otherwise low energy favors local, and remaining cases use cloud. Previous action is passed but ignored during selection, so the rule does not optimize a multi-slot switching objective.

SciPy SLSQP computes shares from

$$\begin{aligned} \min_{\boldsymbol{x}} \quad & -\sum_{i=1}^{N_t} \log(1 + x_i) + 0.01 \sum_i x_i, \\ \text{subject to} \quad & 0.1 \le x_i \le B, \quad \sum_i x_i \le B. \end{aligned} \quad (9)$$

Uniform initialization receives seeded noise of standard deviation 10^{-6}. For generated $N_t \in \{4, \dots, 8\}$ and $B = 20$, the symmetric objective favors equal shares and uses the budget. Missing SciPy or unsuccessful optimization invokes uniform shares. Our reproduction uses the optimizer, preserving the intended path. This is a share subproblem; it does not jointly optimize network and placement against U_t.

For the fixed $B = 20$ domain, every feasible coordinate is below 99, so $\partial f/\partial x_i = -1/(1 + x_i) + 0.01 < 0$. The exact optimum uses the full budget; strict convexity and equal derivatives then yield the unique allocation $x_i^* = 20/N_t$. This is an allocation subproblem, not maximization of the reported network/placement utility. Distributing an unchanged sum differently does not improve that primary score.

Numerical output is approximate. The source supplies no tolerance override or post-solve budget projection. A focused component audit checks all existing seed values 0–4 and100–104 with each $N = 4, \dots, 8$, separately for both versions (100 small solves). Maximum absolute budget residual is 3.1883×10^{-6} and maximum share deviation from the exact optimum is 2.5557×10^{-6}; 26 returned sums exceed the budget strictly. The original exact-budget audit assertion failed and its log is retained. SLSQP reports success while the source accepts small residuals; no exact numerical feasibility is asserted.

At fixed discrete/fidelity/checkpoint choices, these source bounds give $|\Delta u| \le 1.5942 \times 10^{-7}$. Bounding the score derivative by 1.83 gives absolute slot-score impact below 2.918×10^{-7}. The choices themselves do not depend on optimized shares. This conditional bound is too small to reverse either retained negative comparison; it does not repair broader fairness or resource-feasibility limitations. No historical allocation or result is projected or replaced.

4.2 Linear Selector and Greedy Model Evaluator

The stored `information_equivalent` policy scores available network and positive-capacity placement combinations using

$$\begin{aligned} S_t(n, v) = & -0.004\ell_t\alpha_n\beta_v - 0.02q_t \\ & - 0.35\mathbb{1}[\text{outage}, n = \text{terrestrial}] \\ & - 0.002(50 - e_t)\mathbb{1}[e_t < 50, v = \text{cloud}] \\ & + 0.25\mathbb{1}[\text{strict}, v = \text{edge}] \\ & + 0.15\mathbb{1}[\ell_t > 40, v = \text{edge}]. \end{aligned} \quad (10)$$

It chooses the largest score, resolves ties by iteration order, and uses uniform shares and unit power. Loss contributes a constant across candidates, so it does not change the selected action. Switching cost is absent from selection. We preserve all weights without further tuning.

The stored `oracle` evaluates U_t over current available networks and positive-capacity placements with uniform shares and unit power. It considers the previous action greedily. It does not search continuous allocations or plan an episode. Another policy can outperform it over an episode through a different transition path. We call it the *greedy model evaluator*, not an upper bound. An interval spanning zero against it is inconclusive; it is not proof of equivalence or optimality.

4.3 Linear-Selector Collapse and Its Domain

Let $B_e = 0.25\mathbb{1}[\text{strict}] + 0.15\mathbb{1}[\ell > 40]$ and $P_c = 0.002\max\{50 - e, 0\}$. For a fixed available network n, the implemented score differences are

$$S(n, e) - S(n, c) = 0.004\ell\alpha_n(0.15) + B_e + P_c, \quad (11)$$
$$S(n, e) - S(n, l) = 0.004\ell\alpha_n(0.25) + B_e. \quad (12)$$

Here e, c, l in score arguments denote edge, cloud and local placement. If edge is a candidate, $\ell > 0$, $\alpha_n > 0$, and the frozen nonnegative bonuses and placement coefficients apply, both differences are strictly positive. Edge therefore dominates the other placements. These conditions hold for every full original family; independently inspected revised full-family inputs have the same property. They are sufficient conditions, not a theorem for arbitrary weights, absent edge or all interventions.

Loss adds a constant across selection candidates. An outage removes terrestrial from the generated available set, so its mismatch penalty cannot distinguish the remaining candidates. With edge selected, the argmax minimizes α_n among available paths: terrestrial, otherwise local-edge Wi-Fi if available, otherwise degraded-local. The revised optional offline category has coefficient10 and does not beat these available alternatives. Shares are uniform, power is one; the revised linear action defaults to target fidelity and no checkpoint or recovery. Its linear score has no fidelity-choice dimension. Features change score magnitudes or margins without changing this argmax within the stated domain; latency and loss still affect the subsequently evaluated utility.

A focused source check reconstructs the existing five-seed input sequences for train, held-out and three shifted families, preserving each version's random-draw schedule. It calls only the respective selector and independent simplified-rule/inequality assertions. All 9,600 states in each version pass separately. No per-slot action traces were retained in the original summary files; these are reconstructed-input assertions, not a new held-out performance study or additional seed replicates.

The strict result excludes zero latency. Under `no_edge_observations`, $\ell = q = 0$. With non-strict class and $e \geq 50$, all candidate scores tie. Enumeration starts with cloud and the first available network, and replacement uses strict greater-than, preserving that first candidate. Strictness or low energy can change the remaining ties. A new boundary assertion demonstrates this exception. Historical ablation outputs evaluate the procedural policy; the linear boundary check is not an archived linear-policy ablation outcome. Removing edge also removes the dominance premise.

4.4 Actual Baseline Semantics

Table 1 translates stored labels into behavior. Cloud fallback `no_adaptation` changes network when terrestrial is unavailable, contradicting the protocol's "never switches" description. `static` freezes the entire first-slot action, including its shares; later user-count changes also prevent reading it as a fresh per-user allocation. Original JSON labels are preserved for provenance.

4.5 Information and Computational Fairness

The original non-static policies receive the full `Slot`. Equal accessible objects do not imply equal used features. The procedural rule reads edge capacity and class; the linear selector builds a broader vector, including loss as a constant term. The original comparison has no runtime mask. The revised code restricts fields through `PolicyObservation` and separates `PolicyActionSpace`. Procedural declarations include latency, outage, energy, edge capacity, and strictness; linear declarations include latency, loss, outage, energy, and strictness. Edge capacity remains accessible through action-space feasibility. This protects declared access but does not establish identical observation use.

Neither version equalizes computation or optimization opportunities. Procedural and optimization policies invoke SLSQP, the linear rule enumerates discrete actions, and the greedy evaluator scores candidates. Common charges are applied after selection, but most policies do not optimize them. Recorded host execution time is platform-sensitive and excludes sensing, communication, and actuation. No real-time RAN or speed-superiority claim is made.

5 Experimental Design and Reproduction

5.1 Preserved Versions and Actual Sample Size

The original source is `86a5a6f`; its protocol predecessor is `72044d4`. The revised extension is `b44357f`. Full identifiers are in the reproduction supplement. The protocol declares a frozen date of August 18, 2026. The predecessor establishes local ordering before the result-bearing revision. This is a repository freeze, not external preregistration. Paired calculations and charge rescaling here are retrospective additions.

Both versions use train seeds 0–4 and held-out seeds 100–104, 24 episodes per seed, and 16 slots per episode. Each family contains 120 episodes and 1920 slots but only five seed-level replicate values. Paper II has *five seeds*, irrespective of four-seed or 30-seed studies in other materials; incompatible studies are not pooled. Held-out and train families use the same distribution. Their difference is pseudo-random draws, not a new population or a learned policy's generalization.

Table 2 gives major draws. Blockage ranges are 0–0.3 normally, 0.45–0.95 for high blockage, and 0.1–0.6 for outage heavy. Edge-capacity ranges are 0.4–1.0 normally, 0.2–0.8 for high blockage, and 0.3–0.9 for outage heavy. Terrestrial is removed during outage and Wi-Fi

Table 1: Original policy semantics. All are analytical simulations.

Artifact label	Enforced behavior and consequence
`no_adaptation`	Cloud; terrestrial when available, then first available network; uniform shares. Fallback can switch.
`static`	First-slot rule-based action frozen. Later unavailable networks remain scored; no feasible-service guarantee.
`local_only`, `cloud_only`	Fixed placement with category preference fallback; no task or battery dynamics.
`edge_only`	Edge placement; Wi-Fi/degraded-local preference; priority shares. Share fairness is not service fairness.
`rule_based`	Outage-aware network; edge above 60 ms with positive capacity; priority shares.
`optimization_based`	SLSQP shares; preferred available terrestrial/Wi-Fi/NTN network; cloud placement.
`twin_informed`	Threshold network/placement; SLSQP shares; low-energy power reduction; previous action ignored.
`information_equivalent`	Frozen linear scores; uniform shares; no guarantee of equal consumed features or computation.
`oracle`	Greedy current-slot metric search with uniform shares; no global bound or future access.

Table 2: Original synthetic priors. Uniform draws are independent across slots except transition scoring.

Family	ℓ (ms)	q (%)	p_{out}	e
Train/held-out	20–80	0.1–4	0.12	60–140
High blockage	40–140	3–18	0.20	60–140
Energy constrained	20–80	0.1–4	0.12	12–40
Outage heavy	30–120	1–10	0.55	50–120

when edge capacity is at most 0.25. User counts are discrete uniform 4–8, priorities 1–3, and strict class probability 0.25. Cloud/local capacities are 1.0/0.6. High blockage changes several priors jointly, rather than isolating blockage causally.

The revised generator also draws task progress and recovery budget and may append offline continuation. Added draws change subsequent samples even for matching seed integers. Its target-fidelity score increases latency by 5%. Cross-version differences cannot isolate one feature's effect; each version is reproduced separately.

5.2 Statistical Unit and Verification

Utility is averaged over slots, episodes, then seeds. For paired policies A, B, let $d_s = \bar{U}_{A,s} - \bar{U}_{B,s}$. We compute

$$\bar{d} \pm t_{0.975,4}\frac{s_d}{\sqrt{5}}, \qquad s_d^2 = \frac{1}{4}\sum_{s=1}^{5}(d_s - \bar{d})^2. \tag{13}$$

Pairing is valid because policies in a version/family share pre-generated episodes. Five seed means, rather than 1920 slot comparisons, determine inferential sample size. Normality and representativeness cannot be assessed well at $n = 5$. Intervals describe uncertainty conditional on configured priors; they do not describe physical-network uncertainty. Exploratory comparisons and shifted families are not multiplicity-adjusted confirmatory claims. Archived pooled Cohen's d is retained but not presented as paired effect; paired $d_z = \bar{d}/s_d$ is supplied separately.

Original summaries use rounded $t = 2.776$; new tables use SciPy's quantile, explaining small rounding differences in marginal intervals. The recovered historical source reruns used Python 3.12.14, NumPy 2.3.5 and SciPy 1.18.1. The new focused audit uses an existing Python 3.14.7 environment; its exact dependencies are retained separately. Comparable non-timing JSON matches exactly in both versions. The recovered historical records report five original and 31 revised passing tests; those tests were not re-executed merely to relabel them as new validation. New focused arithmetic checks verify 1,477 non-timing summaries, selector assertions and optimizer component behavior; their separate scripts and logs are supplied. Host-process time is excluded from equality because scheduling and platform state are not deterministic outputs. Checksums, source snapshots, reproduction commands, and independent arithmetic are supplied. Reproduction was performed within this author-review workflow; no external laboratory validation is claimed.

6 Results

6.1 The Original Held-Out Hypothesis Fails

Table 3 and Fig. 1 show the original result. Procedural twin utility is 0.79302, below cloud fallback's 0.80662. Their paired difference is -0.01360 with interval $[-0.01478, -0.01242]$. Every seed difference is negative. The original superiority direction fails in this synthetic comparison. It does not establish that twin information is harmful, because rule structure and optimization opportunities are uncontrolled.

The linear selector averages 0.82739. Its paired advantage is 0.02077 over cloud fallback and 0.03437 over the procedural rule; the latter interval is $[0.03326, 0.03548]$. Greedy evaluator utility is 0.82708. Linear minus greedy has interval $[-0.00158, 0.00219]$, supporting neither a detected difference nor practical equivalence. The slightly higher linear mean is consistent with a transition problem whose greedy reference is not globally optimal.

Static and edge-only results require different interpretations. Static can use unavailable actions and supplies an optimistic reference under incomplete feasibil-

Table 3: Original held-out results, five seeds. $\widehat{L}$ is analytical and C is mean episode sum. See Table 1 for label meanings.

Policy	$\bar{U}$	95% seed CI	$\widehat{L}$ (ms)	C / episode
no_adaptation	0.8066	[0.8036, 0.8097]	81.64	0.1533
static	0.8236	[0.8211, 0.8261]	77.86	0.0000
local_only	0.6915	[0.6870, 0.6961]	140.30	0.0000
cloud_only	0.7838	[0.7780, 0.7897]	92.32	0.1533
edge_only	0.8231	[0.8207, 0.8256]	77.17	0.0000
rule_based	0.8041	[0.8015, 0.8067]	76.64	0.3496
optimization_based	0.8066	[0.8036, 0.8097]	81.64	0.1533
twin_informed	0.7930	[0.7904, 0.7957]	77.49	0.4818
information_equivalent	0.8274	[0.8247, 0.8301]	70.99	0.1533
oracle	0.8271	[0.8248, 0.8294]	74.31	0.0392

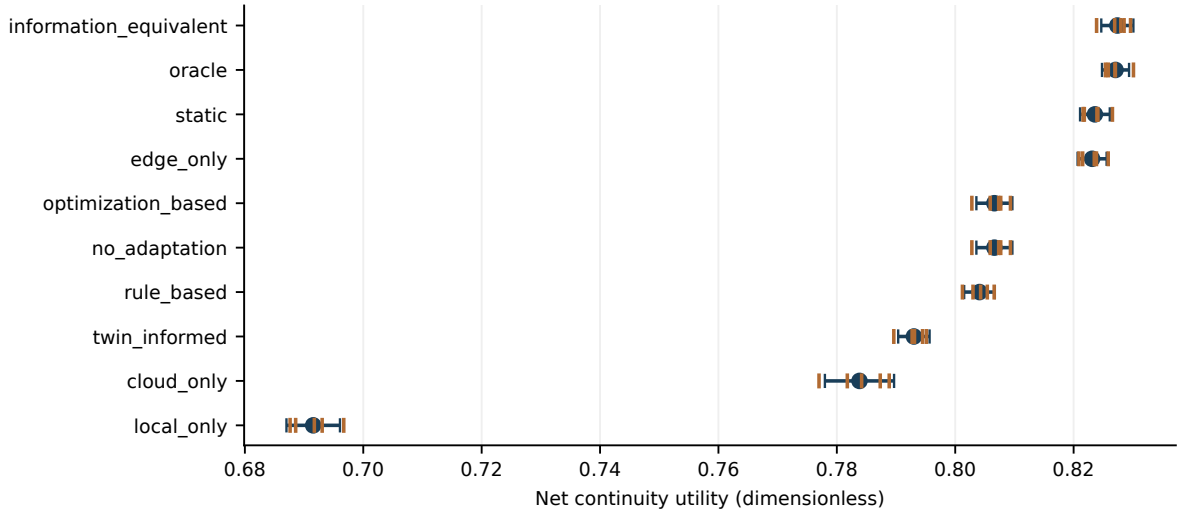


Figure 1: Original held-out utility with 95% seed intervals and individual seed means. Static feasibility is not enforced, limiting its apparent advantage.

ity scoring. Edge-only is a categorical rule rather than physical low-latency deployment. Cloud fallback and optimization-based scores essentially coincide because share optimization does not improve the aggregate-share term. This reflects objective symmetry and scoring architecture, not proof that radio optimization is unnecessary.

6.2 Switching Reverses the Gross Benefit

Figure 2 decomposes net utility and average slot charge. Procedural gross continuity is approximately 0.82313 versus 0.81620 for cloud fallback. Episode charges are 0.48183 versus 0.15333, or 0.03011 versus 0.00958 per slot. A gross gain of about 0.00693 is outweighed by 0.02053 additional slot charge. Threshold-based network and placement decisions ignore the previous action during selection.

Zeroing scored switch penalties with unchanged rule and episodes improves utility by 0.03011, with paired interval $[0.02865, 0.03158]$. It does not remove switching events or estimate physical cost. Recorded charges remain in JSON while subtraction is disabled; the intervention confirms an accounting effect.

For fixed traces, rescale charges by λ:

$$\bar{U}_j(\lambda) = \bar{G}_j - \lambda \bar{C}_j / T. \tag{14}$$

The clipping bound is inactive on these original traces over the plotted range. Equating procedural and cloud-fallback means gives $\lambda \approx 0.33766$ (Fig. 3). This is post-hoc algebra, not a preregistered sweep or a cost-aware controller at each setting. The crossing reduces configured penalties to about one third. Recomputed actions could give a different crossing; no adaptive robustness threshold is established.

6.3 Shifted Families Do Not Establish Uniform Superiority

High-blockage procedural utility is 0.61591 versus cloud fallback 0.59716; outage-heavy values are 0.70463 versus 0.67090. The linear selector is 0.63104 in high blockage and 0.70186 in outage heavy. These descriptive exploratory outcomes establish no uniform superiority (Fig. effig:shift).

Energy-constrained cloud fallback and linear utility retain their held-out values because energy does not directly enter continuity and does not change their actions here. Procedural utility drops to 0.79096 through

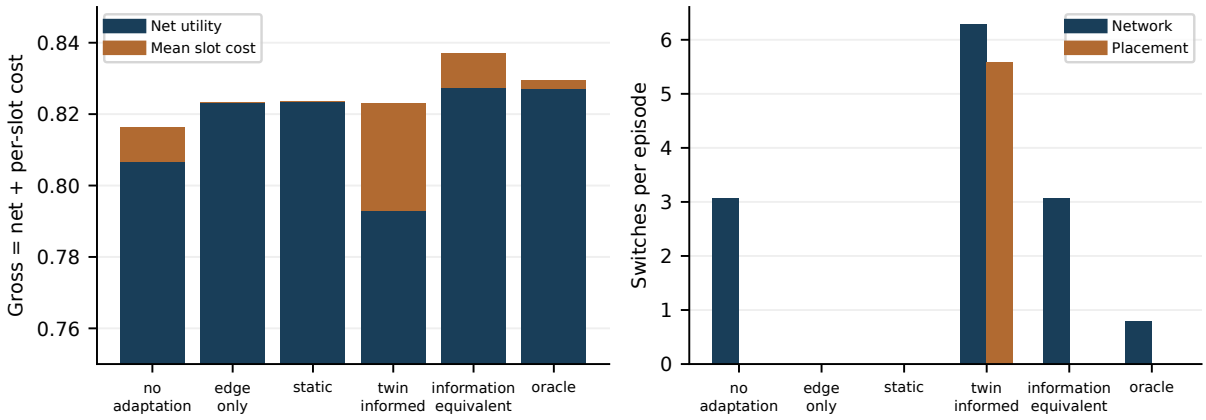


Figure 2: Original switching decomposition. Left: mean slot cost added to net utility recovers gross continuity. Right: switches per episode. Left vertical axis begins at0.75. Charges are configured dimensionless penalties.

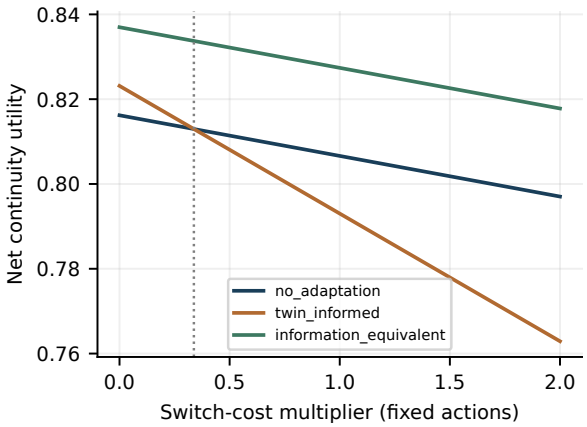


Figure 3: Retrospective cost rescaling on fixed original traces. Vertical mark: procedural/cloud-fallback mean crossing. Aggregate score accounting with fixed actions, not a new sweep or measured transition cost.

its low-energy placement branch. This does not demonstrate energy-constrained service feasibility. A useful energy study requires battery state, task demand, power-duration accounting, and failure on exhausted budget.

6.4 Interventions Reveal Scoring Confounds

Table 4 preserves all original interventions. `no_edge_observations` sets latency and loss to zero in the state used for both control and scoring. Its 0.97971 score reflects a changed scored environment, not evidence that removing observations improves control. `no_energy_constraint` sets the covariate to 10^9, creating enormous energy proxies while leaving original held-out utility unchanged. No enforced battery constraint exists to remove.

`no_twin_context` zeros mobility/blockage and changes strictness; `no_continuity_objective` changes class while keeping (6). Their identical utility follows consumed branches, not removal of the mathematical objective. `no_local_edge` changes availability/capacity. Only disabled subtraction answers a clean narrow accounting question. Observation ablations should preserve a truth state for scoring and provide separately masked controller views.

6.5 The Revised Extension Is Separate and Also Unfavorable

Table 5 reports the revised run. Target, degraded, and minimum-useful fidelity have continuity factors 1.0/0.85/0.65 and energy factors 1.15/1.0/0.75; fidelity changes cost 0.04. Target latency is multiplied by 1.05, minimum-useful latency by 0.9. Checkpoint/recovery charges are 0.03/0.06; checkpoints become stale at eight slots, with further synthetic failure and recovery effects. Availability and age evolve causally during episodes. Progress and recovery budgets remain draws.

Adaptive fidelity averages 0.74198 versus fixed target 0.79650. Adaptive checkpoint averages 0.62375 versus checkpoint disabled 0.68230. Both are unfavorable adaptive outcomes. Paired differences are -0.05452 with 95% interval $[-0.05866, -0.05039]$, and -0.05855 with interval $[-0.06015, -0.05695]$, respectively. These additional paired calculations are post hoc. Base network/placement choices also differ, making these whole-policy comparisons rather than isolated feature effects. No application process checkpoints or resumes. Retrospective reproduction does not turn the extension into the original hypothesis or field evidence.

7 Discussion and Threats to Validity

7.1 What the Negative Result Explains

Threshold adaptation can earn a small gross benefit while frequent transitions dominate utility. Fixed-trace decomposition, source inspection, and paired results support this specific failure mode. It is weaker than a general twin-control claim, but explains more than

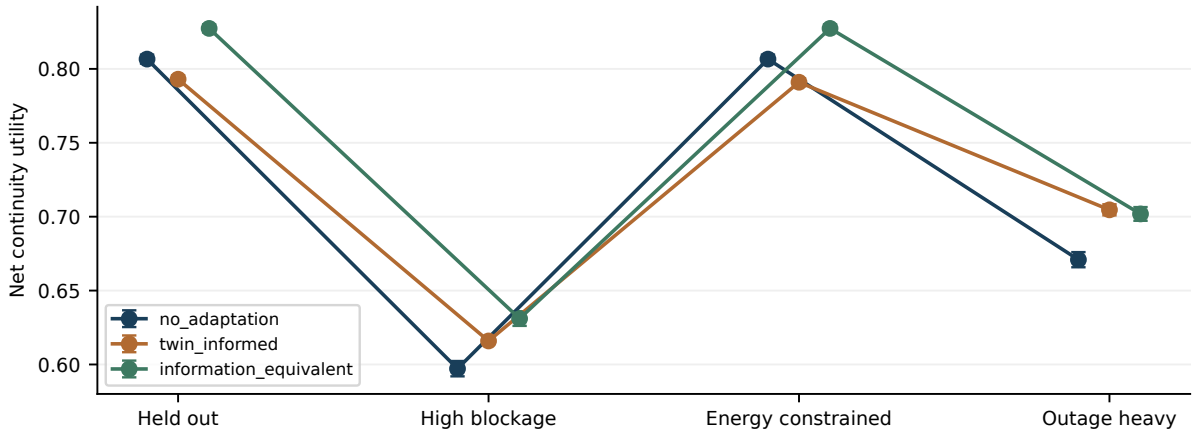


Figure 4: Original family comparisons with five-seed intervals. Multiple priors change together. Equal energy-constrained/cloud-fallback values reflect model insensitivity rather than demonstrated energy resilience.

Table 4: Original procedural interventions. Several change scoring inputs and are not isolated observation ablations.

Intervention	$\bar{U}$	95% seed CI	C / episode
full	0.7930	[0.7904, 0.7957]	0.4818
no_edge_observations	0.9797	[0.9782, 0.9812]	0.3246
no_twin_context	0.7898	[0.7877, 0.7920]	0.5161
no_continuity_objective	0.7898	[0.7877, 0.7920]	0.5161
no_energy_constraint	0.7930	[0.7904, 0.7957]	0.4818
no_local_edge	0.7913	[0.7873, 0.7952]	0.3246
no_switch_cost	0.8231	[0.8205, 0.8257]	0.4818

Table 5: Revised held-out results, five seeds. Do not pool with Table 3.

Policy	$\bar{U}$	95% seed CI
no_adaptation	0.7965	[0.7910, 0.8020]
static	0.8160	[0.8129, 0.8191]
local_only	0.6759	[0.6692, 0.6826]
cloud_only	0.7715	[0.7627, 0.7802]
edge_only	0.8143	[0.8105, 0.8182]
rule_based	0.7941	[0.7885, 0.7997]
optimization_based	0.7965	[0.7910, 0.8020]
twin_informed	0.7840	[0.7805, 0.7874]
information_equivalent	0.8185	[0.8135, 0.8235]
oracle	0.8185	[0.8140, 0.8231]
fixed_target_fidelity	0.7965	[0.7910, 0.8020]
adaptive_fidelity	0.7420	[0.7337, 0.7503]
checkpoint_disabled	0.6823	[0.6776, 0.6870]
adaptive_checkpoint	0.6237	[0.6184, 0.6291]

a lower mean. It motivates transition-aware decisions while leaving physical transition magnitude open.

The model constrains the lesson: shares affect continuity only through their sum, energy is absent from primary scoring, mobility/blockage are unused, capacity is not a queue, and static actions may remain scored through outage. These are material scientific properties. Reproduction verifies fidelity to the program; it cannot verify realism the program lacks.

7.2 Internal, Construct, and External Validity

Code and protocol disagree on labels. Common objects in the original and restricted declarations in the extension provide different access evidence, neither proving equal computation or feature use. Some interventions alter truth. Five seed means limit precision and tail analysis. Narrow intervals quantify randomness conditional on priors, not uncertainty about real networks.

Utility is a smooth reliability-delay product rather than completed-task fraction, application persistence, or a service-specific minimum-useful threshold. Share fairness, predicted delay, and continuity partly reuse analytical quantities, so multiple metrics do not give independent validation. Energy has unsupported physical units. Host timing is not air-interface or end-to-end latency. NTN is a multiplier without geometry, shared satellite capacity, or terminal measurements.

The companion ReadyGary project uses synthetic 28 GHz inputs and host-process timing; those are not integrated evidence for the orchestration score. SpectrumX judged IQ evaluation remains separate and unmodified. Neither validates (3)–(8). No university affiliation, supervision, or field campaign follows from a roadmap.

7.3 Prospective Fair Comparison and Physical Validation

A new study should freeze a new identifier before outcomes, preserving both archives. It should separate truth from observation, give comparable policies identical declared views, enforce feasible actions, and impose an explicit compute budget. Hysteresis or transition-aware discrete optimization should use the same avail-

able action set. A finite-horizon dynamic program could supply a true optimum for the small discrete model, remaining explicitly privileged.

The protocol should distinguish ordinary seeds from shifted priors, select seed count through precision/power analysis, and specify one primary paired contrast and multiplicity treatment. Costs should be swept with actions recomputed. Causal energy and task state should be modeled, or claims restricted to score accounting. No new experiment is inserted into old results to repair a failed hypothesis.

Physical validation requires measured latency/loss, unavailable-action behavior, migration interruption, target-device energy, task outcomes, and state-estimation error. These would calibrate and test the predictor. They are future studies, not completed evidence.

8 Conclusion

Exact reproduction retains unfavorable procedural-twin results and unfavorable adaptive results in a separate extension. Original switching charges outweigh a small gross-continuity advantage. A frozen linear rule does better, but unequal policy semantics, incomplete feasibility, and unequal computational work prevent attribution to twin information alone. The selector characterization and numerical audit further show where apparently rich decisions reduce to coefficient ordering and an approximately uniform subproblem. The study provides a transparent, implementation-specific reproducibility case, with no claim of established novelty, and a prospective path to a fairer comparison.

Data, Code, and Author-Review Declarations

The separate bounded evidence archive includes both frozen source/protocol vintages, original and reproduced summaries, seeds, checksums, independent mathematical/statistical scripts and environment records. Exact revisions are supplied in `PROVENANCE.json`; original 86a5a6f, revised b44357f and manuscript/preprint revisions are distinct. No permanent evidence DOI or public preprint record has been created. Repository software MIT notices are retained; article/data permissions and redistribution rights require owner confirmation. Existing journal-review files remain preserved separately.

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AI-Use Disclosure

OpenAI ChatGPT/Codex tools assisted substantively with drafting/revision, source and policy audits, mathematical exposition, independent statistical and assertion code, literature/metadata investigation, programmatic plots, LaTeX and reproducibility packaging. The abstract, Sections 1–8, equations, code, figures and declarations include AI-assisted material. Plots are rendered from synthetic artifacts, not invented measured data. The historical tool/model inventory remains subject to owner confirmation. AI systems are not authors. Edmund Gunn, Jr. must verify the complete text, methods, computations, citations and rights before any release. Human scientific approval, originality verification, article-license approval and publication authorization remain false. No independent human peer review is claimed.

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